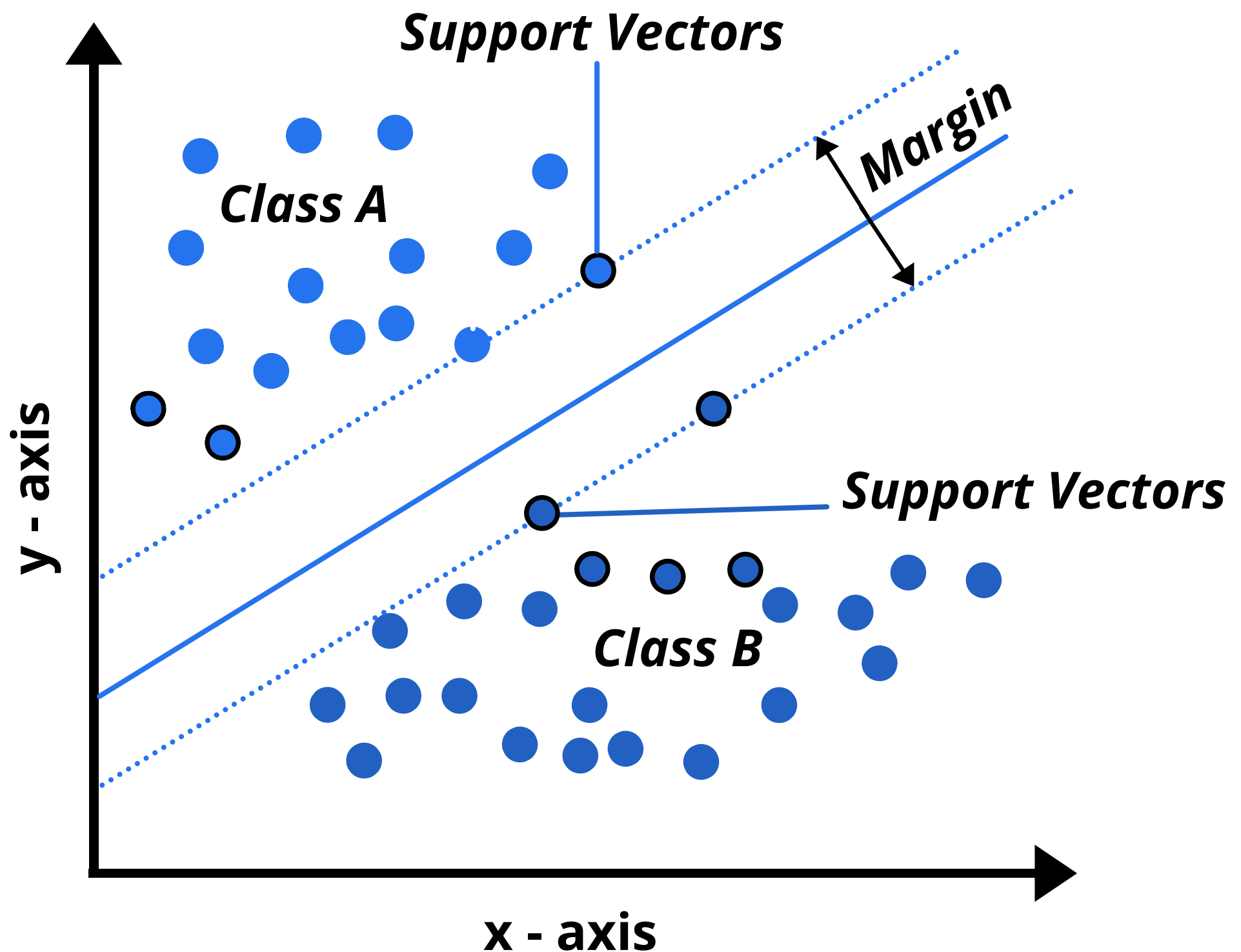


40+ Most Asked

Interview Questions on

SVM



- What is a Support Vector Machine (SVM)?
- How does SVM differ from other classification algorithms?
- What are support vectors in SVM?
- Can SVMs be used for both classification and regression tasks?
- What is the role of the kernel in an SVM?
- Explain the concept of the hyperplane in SVM.
- What is the margin in an SVM model?
- How does SVM handle linearly non-separable data?
- What are the advantages of using SVM over other classifiers?
- Can SVM be used for multi-class classification? If yes, how?
- Explain the concept of the kernel trick in SVM.
- What is the mathematical formulation of a linear SVM classifier?
- Describe the Lagrange multipliers in the context of SVM.



- How does SVM minimize the structural risk?
- What is the primal and dual formulation in SVM?
- Explain the concept of slack variables in SVM.
- How does the choice of kernel affect the SVM model?
- What is the significance of the regularization parameter in SVM?
- Describe the hinge loss function in SVM.
- How is the decision function formulated in SVM?
- What are common kernel functions used in SVM?
- How do you select the right kernel for an SVM model?
- What are the steps involved in training an SVM model?
- How does feature scaling affect SVM performance?
- Describe how to use SVM for a text classification problem.
- What are some common challenges in training SVM models?



- How do you handle imbalanced classes in SVM?
- What techniques can be used for parameter tuning in SVM?
- Can SVMs handle large datasets efficiently?
- How do you interpret the results of an SVM model?
- Explain the concept of Sequential Minimal Optimization (SMO) in SVM.
- What is the role of the C parameter in SVM?
- How does the gamma parameter in the RBF kernel affect the SVM model?
- Discuss the concept of feature mapping in relation to SVM kernels.
- How do SVMs perform in high-dimensional spaces?
- Explain the concept of soft margin in SVM.
- What are some modifications to standard SVM for handling large-scale data?
- Discuss the use of SVM in anomaly detection.
- How can SVM be combined with other machine learning techniques?



- What are recent advancements in SVM research?
- Provide an example where SVM outperforms other classification algorithms.
- How is SVM used in image recognition tasks?
- Discuss the application of SVM in bioinformatics.
- How can SVM be applied in financial market prediction?
- What role does SVM play in sentiment analysis?